

Q1: 24 Feb (Shift 1) - Numerical

The coordination number of an atom in a body-centered cubic structure is ____ [Assume that the lattice is made up of atoms]

Q2: 25 Feb (Shift 2) - Numerical

The unit cell of copper corresponds to a face centered cube of edge length $3.596\text{\AA}$ with one copper atom at each lattice point. The calculated density of copper in kg/m^3 is ____ [Molar mass of Cu: 63.54 g; Avogadro number $= 6.022 \times 10^{23}$]

Q3: 26 Feb (Shift 2) - Numerical

The number of octahedral voids per lattice site in a lattice is ____.

(Rounded off to the nearest integer)

Answer Key**Q1 (8)****Q2 (9077)****Q3 (1)**

Hints and Solutions

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Q1 (8)

Fact

Q2 (9077)

$$a = 3.596 \text{ \AA}$$

$$d = \frac{Z \times GMM}{N_A \times a^3}$$

$$d = \frac{4 \times 63.54 \times 10^{-3}}{6.022 \times 10^{23} \times (3.596 \times 10^{-10})^3}$$

$$d = 0.9076 \times 10^4 = 9076.2 \text{ kg/m}^3$$

Q3 (1)

Assuming FCC

No of lattice sites = 6 face centre + 8 corner = 14

No. of octahedral voids = 13

$$\text{Ratio} = \frac{13}{14} = 0.92857 = 1 \text{ (Nearest integer)}$$